Proceedings of 7th International Conference ITELMS'2012

Opportunities of Adaptive Control Algorithms Application in Railway Control Systems

A. Potapovs*, M. Gorobetz**, A. Levchenkov***

** Institute of Industrial Electronics and Electrical Engineering, Riga Technical University, 1, Kalku Str., LV-1658, Riga, Latvia, Andrejs.Potapovs@rtu.lv*
*** Institute of Industrial Electronics and Electrical Engineering, Riga Technical University, 1, Kalku Str., LV-1658, Riga, Latvia, mihails.gorobecs@rtu.lv*
**** Institute of Industrial Electronics and Electrical Engineering, Riga Technical University, 1, Kalku Str., LV-1658, Riga, Latvia, anatolijs.levcenkovs@rtu.lv*

Abstract

The authors of the article on the example of a particular crossing suggest a description of a new project of adaptive control system of railway crossings work selecting for the adaptive system the most suitable adaptive control algorithm defining its main blocks, their functions, and operation algorithms.
At the beginning of the paper the essence of complex objects control adaptation is analysed, and the most optimal one is selected from them. Then the main blocks of the suggested system and their operation are described applying the selected adaptive functioning algorithm. The paper is completed with the comparison of the effectiveness of the suggested system with the present one, proving that the suggested system can significantly compete and solve the problems with traffic in the movement of railway and sity transport.

KEY WORDS: *railway transport; embedded systems; adaptive control algorithm*.

1. Introduction

Analysing the infrastructure of the present transport movement regulation the authors have come to a conclusion that the mutual regulation of the railway and city transport flows faces the problems becoming apparent in the traffic congestions in city or limited movement of the railway transport [1]. All these factors result in drawbacks for both types of the transport as well as increase time of movement for people within the range of particular routes.

A part of these problems is formed by the ageing automation system having no opportunities to adapt to the increasing transport flow. It causes the necessity to develop a new adaptive system of railway crossings applying adaptive control algorithms being able to solve the considered problem, increasing the carrying capacity of both transport types flows adjusting to the movement of railway transport. Developing a new adaptive control system it is important not to complicate a lot the existed control system for maintaining the present level of safety.

The authors make an assumption that a significant decreasing of city transport idle stay time can be reached improving the control systems of the crossings with the programmable logic controllers (PLC), the control programs of which apply adaptive algorithms, and specific sensors for giving an information about the position of a train.

2. Purpose and tasks

The purpose of the authors is to develop a new project of crossings control system applying adaptive control algorithms.

The main tasks are:

- To analyse an adapting essence of the complex objects;
- To state the task of the adaptation of crossing control system;
- To define the operation principles of the suggested system and its main blocks;
- To prove the effectiveness of the suggested system by means of numeric example.

3. The essence of the complex objects control adaptation

The process of adaptation allows to realise the alignment of the control system to different specific operation conditions as well as variable conditions of environment. It considers some hierarchic levels [2].

One the most convenient from them is the parametric adaptation. Applying this type of adaptation the parameters of the control system are aligned to a particular case. The application of the adaptation of this type exists in the cases when instability of the variable parameters of the control object is obvious or other emergency situations take place during the operation of control system. The adaptation in this case allows alignment of the control system at each step of its operation not applying as an output the feedback of information flow between the control system and object under this control.

The adaptive control process of such type is often called a control with adaptive model of the object. The obvious advantages are the applied methods and equipment being close to the point of identification. Therefore the adaptation of the parameters is connected with their identification, i.e. determination of the object's parameters in its normal functioning mode.

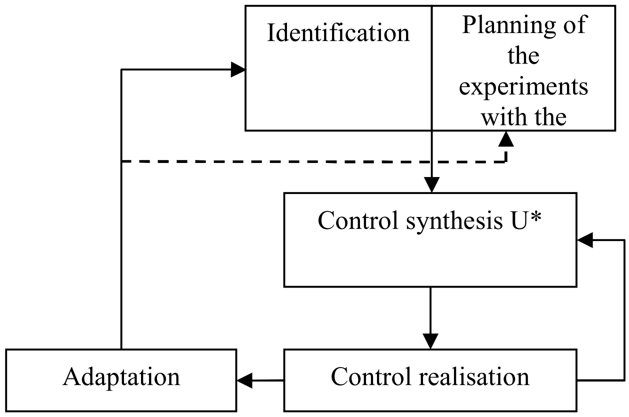


Fig. 1. Order of the control steps of a complex object

However the process of the object control does not often give much enough information on the model correction as the control process is not different enough to inform on the object's specific features necessary for the synthesis of the control process and illustration of the object's model.

This circumstance results in an artificial input of different control signals close to the normal operation conditions. The organization of these signals realises also the following adaptation order demonstrated in fig. 1 with the help of dotted line. Unfortunately in this case the effectiveness of the control is decreased however guaranteeing the successful control at the following steps.

The adaptive control achieving the purposes during its process as well as specifying the parameters of the object is called a dual control as due to its specific organization it is directed to two purposes - object control and adaptation.

4. Statement of the adaptation task

The essence of the adaptation task is to modernise the control system of the railway crossing realising it taking into account the speed of the railway transport. .

Let us consider a particular task of a control system VS of a railway crossing (fig. 2) as a control task. The control system under consideration mutually interacts with automobile and railway transport. The condition of the control system SS depends on the condition of train VS. The condition of the train VS is characterised with such criteria like speed V, coordinate X, length of the train L, etc. The control condition VS can be changed applying its adapting factors U [3]. The most important of these factors is time of delay $t_{aiztures}$ determining in what time the crossing is closed for the autotransport and train can cross the point approaching it.

The purpose of adaptation Z* defines the requirements to the criteria of a particular control system condition. These criteria can be of three types:

1. Criteria of inequality:

$$H(U) = (h_1(U),...,h_p(U)) \geq 0$$

2. Criteria of equality:

$$G(U) = (g_1(U),...,g_s(U)) = 0$$

3. Criteria of minimization:

$$Q(U) = (q_1(U),...,q_l(U)) \longrightarrow \min$$

The purpose of the adaptation task is connected with the solution of the following task:

$$Q(U) \longrightarrow \min \Rightarrow U^*$$

where S: $\begin{cases} H(U) \geq 0; \\ G(U) = 0. \end{cases}$

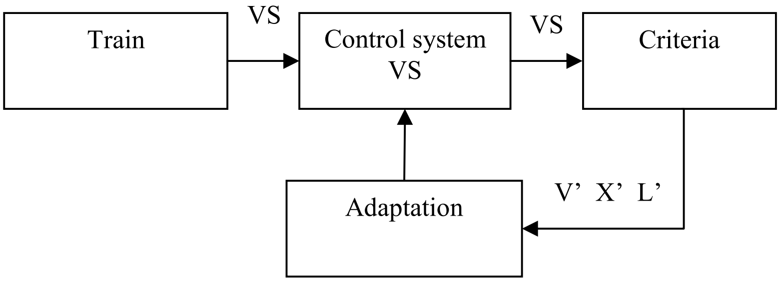


Fig. 2. Adaptive block diagram of the control system

For the solution of this task a criterion of minimization is selected determining the maximum minimization of the autotransport ineffective time of staying $t_{gaidīšanas}$, without braking the existing safety norms of railway transport routing.

One of the simplest algorithms of case searching is selected for the role of searching algorithm. This algorithm is called a case searching algorithm with linear tactics. The case searching of this type is based on the use of two operators. The first of them is case step operator ζ and the second is the operator of the previous step repeating (+). The work of each operator can result in either minimization function Q is decreasing $(\Delta Q < 0)$, or not decreasing $(\Delta Q \geq 0)$. Here $\Delta Q_N = Q_N - Q_{N-1}$, but $Q_N = Q(U_{N-1})$ – value of minimized function at N-th searching step. According to the result one or other operator is activated.

The algorithm of case searching with linear tactics is based on the following obvious assumption on the optimization of the object's parameters: the possibility of success $(\Delta Q < 0)$ in the previous successful direction is higher that selecting this direction at random. In other words it is effective to repeat a successful step, but in the case of failure to make a case step, i.e. turn to operator ζ.

The linearity of the algorithm tactics is based on the imitation of linear behaviour, i.e. direct repetition of a successful step.

This algorithm is easy to illustrate with two oriented graphs, transferring from one operator to another in the case of failure. As it is seen in the successful case - the minimization function is decreasing - the algorithm repeats (+) its step ΔU, resulting in this success. In the case of failure the algorithm turns to the probabilistic operator ζ and repeats it till the moment of realization of condition $(\Delta Q < 0)$, i.e. till the first successful step.

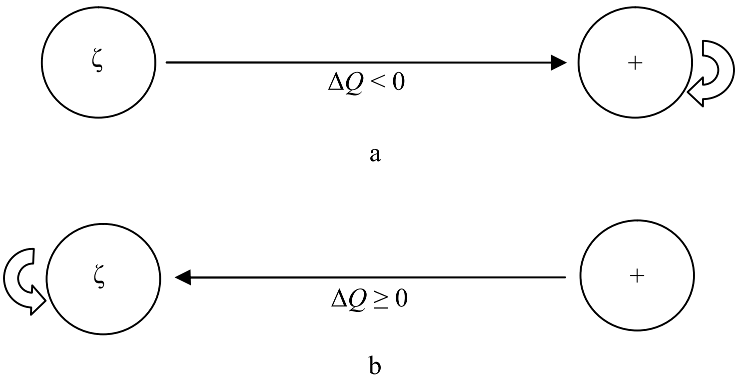


Fig. 3. Graphs of case searching algorithm with linear tactics a) in successful case; b) in the case of failure

In recurrence form this algorithm can be illustrated as follows:

$$U_N = \begin{cases} a\xi \ \text{ when } \ \Delta Q_{N-1} \geq 0 \\ \Delta U_{N-1} \ \text{ when } \ \Delta Q_{N-1} < 0 \end{cases}$$

where a – volume of step ($|\Delta U| = a$); ζ – value ($|\Delta \xi| = 1$) case vector, evenly directed in all directions of the optimized parameters {U}. It means that all directions of vector ζ are equal.

5. Structure of new system of crossing control and its main blocks

Further for the realization of the selected adaptation algorithm a structure of new system is described. The mentioned above task is to develop a new adaptive control system of crossing that can as much as possible improve the traffic capacity of railway crossing for autotransport following all the necessary safety norms.

The suggested adaptive control system consists of 3 main parts:

a) System of sensors installed at the point of railway route and locomotives to control along this route (Fig. 3);
b) Programming logic controller installed in an enclosure of crossing control relay (VRS);
c) Existing control system of the crossing equip0ment.

System of sensors – the first task of it is to identify a train coming to the crossing. Sensors D_1 and D_2 installed at the railway route obtain a signal from sensor D_3, installed at the train. For the case when this point is crossed by some undefined train sensor D_4 is installed for fixing it. If obtaining a signal from D_4 there are no signals from D_1 and D_2 then the crossing is blocked without taking into account closing time $t_{aiztures}$.

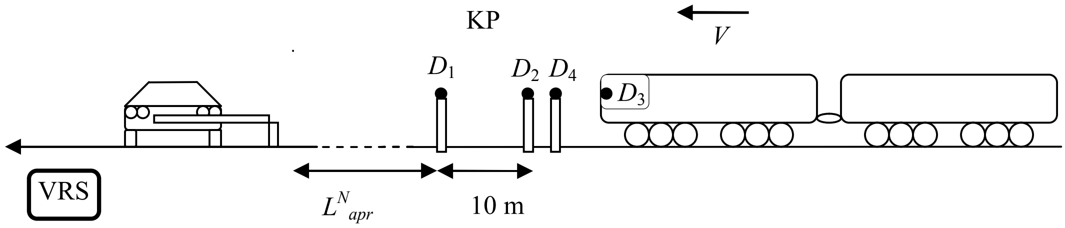
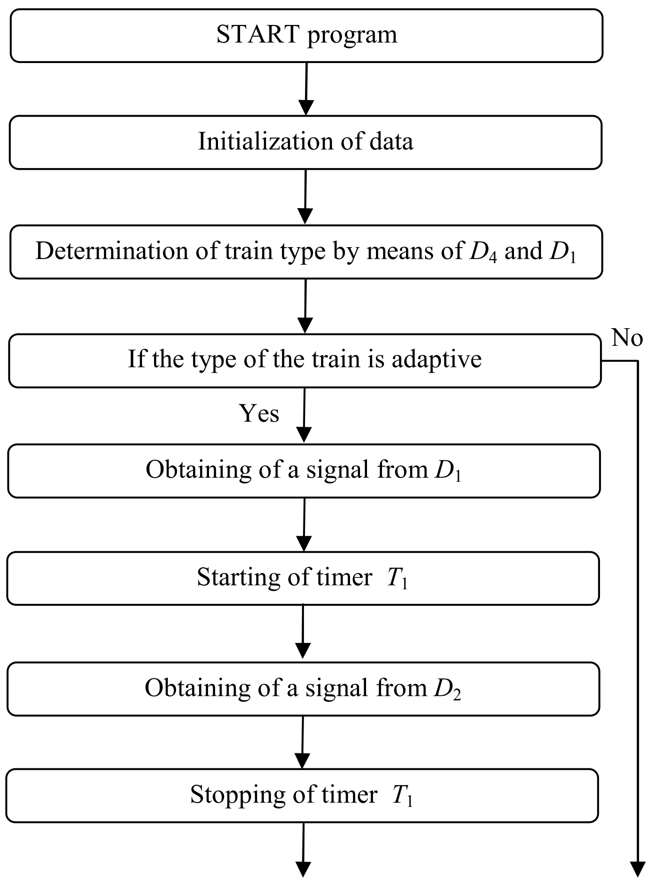


Fig. 3. Principal control scheme of adaptive railway point

The second task of the system is to test whether the train has achieved the preset control point KP. The given control point is situated at the distance $L^N_{apr} = t_{vestijuma} \cdot V$ from the crossing (the close sector), where $t_{vēstījuma}$ is the autotransport message time meeting the safety requirements, but V is the maximum speed of train at the railway sector under consideration.

The third task of the sensor system is to assist in determination of the train speed V. With this purpose the sensors are placed at the distance of 10 m (for example) each from other, by means of that the signals defining the speed of the train are obtained with a fixed time intervals.

The article does not contain the description of the application of the auxiliary sensor block necessary at all the identification railway sector for the determining of any train coming on time to the open crossing. This information will be accessible in further papers.



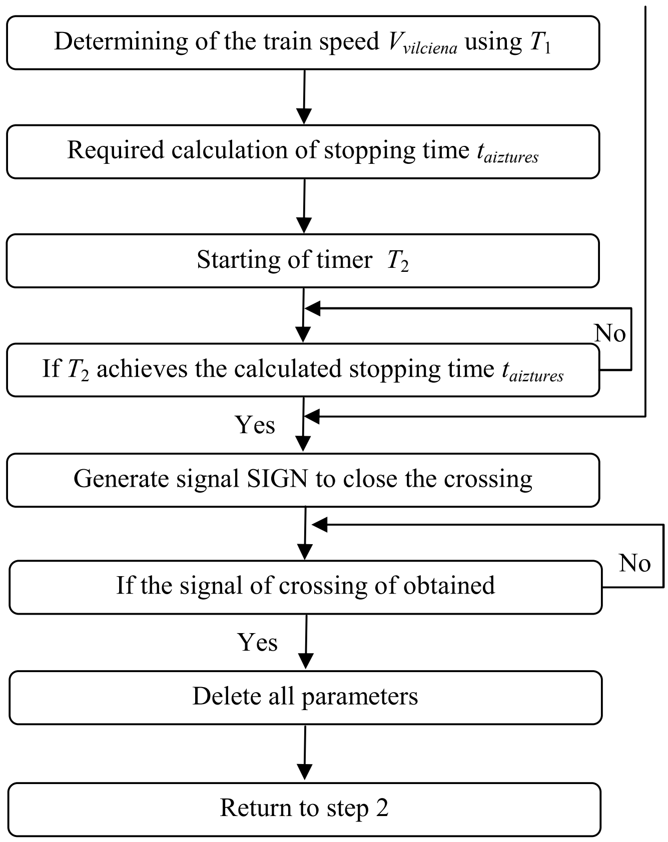


Fig. 4. Block diagram of PLC control program algorithm

The programmable logic controller (PLC) – PLC is installed in an enclosure of crossing control equipment relay (VRS), and its task is to obtain signals from the sensors about the train approaching, make the necessary calculations and generate the control signal for the existing control system of the crossing. The block diagram of the PLC control program algorithm is in Fig. 4.

The present control system of the crossing equipment – this system should control the equipment of the crossing (traffic lights, barriers, etc.) obtaining the control signals from PLC. Having obtained a signal from the existing rail circuit that the train has crossed the point the blocking equipment of the crossing opens the way for autotransport.

6. Example of a new adaptive control system operation

Sensor D_1 installed on the railway obtains a signal from sensor D_3 installed at train and sends it to the PLC installed in the control relay enclosure. In time $t_1 = 2$ s the same signal is obtained by sensor D_2 installed on the way and also sends it to PLC. Getting these signals The PLC control program calculates the speed of the train:

$$V = \frac{10}{2} = 5 \text{ m/s, or } 18 \text{ km/h} \tag{1}$$

At the next step PLC control program calculates duration of the train way to the crossing at the calculated speed that in this case will be equal to $t_{kustības}$:

$$t_{kustibas} = \frac{761}{5} = 152.2 \text{ s} \tag{2}$$

Further using time $t_{vēstījuma} = 34.3$ s, control program of PLC calculates necessary close time $t_{aiztures}$, equal to

$$t_{aiztures} = t_{kustibas} - t_{vestijuma} = 152.2 - 34.3 = 117.9 \text{ s} \tag{3}$$

In fact it is the time that could be saved at the given case if to compare it with the present system of equipment control. The obtained time and safe distance $S_{drošības}$ corresponding to time $t_{vēstījuma}$ at different values of speed are given in Table 1.

Table 1

V_{tek}, km/h	$S_{dr.tek}$, m	$t_{aiztures}$, s
10	95.27	240.09
15	142.91	148.63
18	171.5	118.14
20	190.55	102.89
30	285.83	57.16
40	381.11	34.29
50	476.38	20.57
60	571.66	11.43
70	666.94	4.89
80	762.22	≈0

From Table 1 it is obvious that at the speed of trains of 15-20 km/h that is mostly the speed of the cargo trains it is possible to save 100-150 s at the expense of close crossing, that is large enough period of time, that will give an opportunity to decrease the time of autotransport staying in city.

7. Conclusions

The following conclusions have been made working on the given paper The application of the adaptive algorithms modernising the present railway control systems (crossing control systems) can as much as possible adapt it to the increasing transport flow decreasing staying time of the transport and increasing an average movement speed at the given routes or the intensity of movement.

With the example of a particular project using data of a real crossing where the established problem is topical the following is proved: at low enough values of speed of trains (10-15 km/h) it is possible to decrease the time of inefficient staying of the city transport up to 8 min. This time is saved in the case when the given crossing is crossed by one train but the fact should be taken into account that during twenty-four hours the number of these trains achieves 30-40 units therefore the total time is significantly increases.

The authors are sure that the further investigation of the problematic sectors of the railway system will discover more places for application of the algorithm of the considered type, renovating the old system, its equipment, allowing not only to increase carrying capacity but also to improve the level safety in this system.

References

1. **Altbergs T., Balckars P., Bukovskis V.** un citi. Dzelzceļi. Vispārīgais kurss. Valsts akciju sabiedrība „Latvijas dzelzceļš ”. RIGA, 1996 –
2. **Растригин Л. А.** Адаптация сложных систем. Методы и приложения. Рига, 1981. – 394 lpp.
3. **Charles L. Philips, Royce D.** Harbor Feedback control systems. New Jersey, 1998. – 615. p.
4. **Ribickis L., Ļevčenkovs A., Gorobecs M.** Basics of system theory in the modeling of industrial electronics. RIGA, 2008 – 95 lpp.
5. **Колесов Ю. Б.** Объектно–ориентированное моделирование сложных динамических систем. Санкт-Петербург, 2004 – 232 с.